
CLASSIFIED // OPERATIVE DOCUMENTATION

PROJECT PLAN DOCUMENT

HEAT SIGNATURE

— CLONE PROJECT —

SHIP ROSTER

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Table of Contents

Initial Feature Scope	3
Allocated Tasks	5
Timeline	6
Key	6
Mechanics that are dependent on each other.....	7
Project Structure.....	8
Group Conventions	8
Communication	8
File management.....	9
Repository (GitHub) Management.....	9
Example of the process using GitHub.	10

Initial Feature Scope

This scope outlines and breaks down all the mechanics and features the group initially planned to clone from the original game. Detailed descriptions of the specific gadgets and weapons can be found here.

Note: This reflects the *initial* project scope, as not all these features were ultimately implemented. The reasoning behind these changes is explained in the Game Design Document.

Table 1- A table showing the mechanics that need to be implemented for this clone game

Main Mechanic	Features related to the mechanic that need to be implemented
Character Controller	a. Movement b. Gadget Activation c. Attacking (Melee & Ranged) d. Throwing Items e. Item and Key Pickup
Inventory System → This system will activate when the player pauses time or goes to the shadow map	a. 8 Inventory Slots b. The ability for the user to use slot 1&2 using left and right mouse click input c. User can move freely between inventory slots d. User can pick up items freely
Enemies	a. Patrol Routes within a fixed distance b. Detection of player (a bar that fills up when the enemy gets suspicious and then eventually detects the player, like the eye and yellow to red icon in Assassin's Creed) c. Attacking (Melee & Ranged)
Gadgets	a. Visitor →Teleport that takes you back to the original position after 1 second) b. Sidewinder→ Teleport mechanic that can only teleport you if there is a direct path (no obstacles in between) from the player and destination

	c. Swapper → Swap locations as the player with a selected enemy d. Slipstream → Slows down time by 10x and makes you 5x faster (this feature only lasts one second, which makes it a special ability)
Weapons	a. Longblade → Melee weapon with 8-meter dash, 0.4 second cooldown b. Concussion hammer → Melee weapon with an 8-meter dash and an almost immediate cooldown c. Shortblade → Melee weapon with a 2-meter dash and almost immediate cooldown d. Gun → Standard gun with infinite ammo and one shot -one kill mechanic
Assets	a. Item Sprites b. Player Sprite c. Enemy Sprite d. Map e. Item Effects
Map	a. Standard Doors b. Locked Doors, will only open when the player picks up the corresponding key c. Chests that give the player more items
Shadow Map	a. All doors are locked b. No chests can be opened c. Player cannot use items while using shadow map d. Enemies cannot see the player e. Potential indestructible enemy that chases the player in this map
Camera	a. Follows the player from a 2D top-down perspective. b. Periodically moves rooms

Allocated Tasks

Table 2- A table showing the allocated tasks to each group member

Shreyen	Abolisha	Wilrich
Character Controller → Basic Movement	Character Controller → Gadget Activation → Item and Key Pickup	Character Controller → Improvements and final implementation → Throwing Items
Enemies → Patrol Routes → Melee and Ranged Attacking → Basic Movement	Inventory System	Shadow Map
Map → Doors and Locked Doors → Tilemap of Map for game usage.	Gadgets → Visitor → Sidewinder → Swapper	Slipstream Time Manager
Camera → Follows Player	Documentation → Iterative Process	Weapon System/ Attack → Longblade → Concussion hammer → Shortblade Melee → Gun
Documentation → Finalising both documents → Hypothesis → Game Research: Category and Sub-category Analysis → Iterative Process	Main menu • Creating pause menu And the main menu scene.	Documentation → Design Question → Actions and Challenges Section → Iterative Process

Timeline

This table shows our self-set deadlines. At the end of each day, we showcased what we have done to each other.

Key

Member	Colour
● Shreyen	Red
● Abolisha	Green
● Wilrich	Blue
♥ All Members	Pink

Table 3- A table showing the tasks and deadlines to complete our clone game.

Date	Tasks
Aug 05	Decide Game
Aug 06	Admin and Starting Documentation
Aug 07	Character Movement, Camera, Item system
Aug 08	Tilemap done, All Weapons /Attack System
Aug 09	Inventory Done, Map Doors Done, Character Attack Done
Aug 10	Enemy movement/patrol, Slipstream, Item Pickup Done

Aug 11	Visitor Done, ShadowMap System Done, Enemy player detection/ attacks (no need for time slow)
Aug 12	Playable slice done.
Aug 13	Swapper Done, Map Interactables done.
Aug 14	Bug fix and Playtest
Aug 15	Sidewinder Done
Aug 16	Main menu & buttons, documentation deadline.
Aug 17	Bug fix and Playtest
Aug 18	Game Done, Document Done
Aug 19	Marking Day

Mechanics that are dependent on each other

- Item and key pickup are dependent on the inventory.
- Inventory is dependent on Items.
- Items are dependent on Item Scriptable Objects.
- Attacking is dependent on Weapons and Attack System
- Shadowmap is dependent on the Map.
- Enemy AI is dependent on the tile map.
- Weapons are dependent on Character Item Usage
- Time Manager is dependent on all scripts that use time.
- Throwing weapons is dependent on Item Pickup.

Project Structure

Group Conventions

All group members followed the same coding structure and conventions when creating scripts. This was done to improve consistency and make the project easier to maintain, especially during stages where multiple group members worked on the same parts of the project (for example, the Character Controller). Following this same structure made it easier to understand each other's code and build upon existing systems.

```
//Script name should be what it does (PlayerMovement,InventoryManager).
0 references
public class ExampleScript
{
    //Private variables.
    0 references
    private float __variableName;
    //Public variables.
    0 references
    public float VariableName;
    //Constants.
    0 references
    const float VARIABLENAME = 0F;

    //Functions should be named after what they do (MovePlayer,PickUpItem).
    0 references
    private void FunctionName(float variableName)
    {
        //You don't need to add function before the variable name.
        float functionVariableName;
    }
}
```

Figure 1- A figure showing an example of how all group members' scripts should look.

```
[SerializeField] private GameObject _exampleObject;
private void UseExampleObject()
{
    if(_exampleObject != null)
    {
        _exampleObject.SetActive(true);
    }
}
```

Figure 2- A figure showing an example of a null check that group members should include in their scripts.

Communication

The main form of communication throughout the project was the WhatsApp group chat. This chat was used for general discussions, updates, and coordinating/assigning tasks. Online meetings and project showcases were conducted through Discord, while the integration and merging of content created by each group member was done during in-person meetings.

File management

To keep the project organized, files were placed in specific folders based on their purpose:

Every script should go under Assets -> Scripts.

Every prefab should go under Assets -> Prefabs.

Every sprite should go under Assets -> Sprites.

If there is something new, create a folder and name it after what it should hold. This helped keep the project organized and made it easier for group members to locate necessary files.

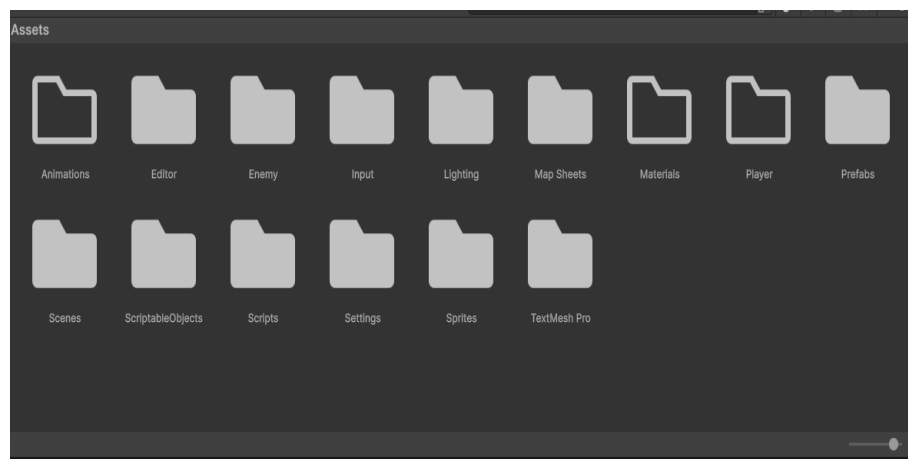


Figure 3- A screenshot of the filing system in the project

Repository (GitHub) Management

When committing changes to the repository, each commit message was written to describe the main changes that the group member completed. If there was insufficient space to describe the work in the commit message fully, additional information was included in the commit description. Branches were named according to the main feature or task being developed within that branch. This made merging content much easier in the development process.

Added script folder and example script Acriid • just now	Default branch	
	main	35 minutes ago
	Recent branches	
	Gadgets	35 minutes ago
Initialize gitattributes Acriid • 12 hours ago	Assets	35 minutes ago
Initial unity project creation Acriid • 12 hours ago	Inventory	35 minutes ago
	Weapons	35 minutes ago
	Enemies	35 minutes ago
Initial commit Acriid • 20 hours ago	Other branches	
	CharacterController	35 minutes ago
	Map	35 minutes ago

Figure 4.1 & 4.2 - How our repository was structured

